

PINE SPRINGS GUADALUPE PASS, TX, USA

WMO: 722620

Lat: 31.831N

Lon: 104.807W

Elev: 1663

StdP: 82.88

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23055

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 12	(b) -7.0	(c) -4.6	(d) -20.9	(e) 0.7	(f) 10.3	(g) -18.5	(h) 0.9	(i) 8.9	(j) 22.7	(k) 6.9	(l) 21.3	(m) 5.8	(n) 11.2	(o) 60	(p) 0.817

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 6	(b) 12.8	(c) 35.2	(d) 15.6	(e) 33.7	(f) 15.7	(g) 32.3	(h) 15.6	(i) 19.4	(j) 27.4	(k) 18.8	(l) 26.9	(m) 18.2	(n) 26.3	(o) 7.5	(p) 250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
17.4	15.3	20.4	16.8	14.7	20.4	16.1	14.0	20.6	62.4	27.3	60.3	26.6	58.3	26.0	21.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 21.1	(b) 18.9	(c) 17.0	(e) -9.7	(f) 38.1	(g) 2.5	(h) 1.0	(i) -11.5	(j) 38.8	(k) -13.0	(l) 39.3	(m) -14.4	(n) 39.9	(o) -16.2	(p) 40.6		
(4) 21.1	(5) 18.9	(6) 17.0	(7) -9.7	(8) 38.1	(9) 2.5	(10) 1.0	(11) -11.5	(12) 38.8	(13) -13.0	(14) 39.3	(15) -14.4	(16) 39.9	(17) -16.2	(18) 40.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	16.2	6.2	8.7	12.2	16.4	20.9	25.4	25.2	24.6	21.4	16.7	10.7	5.9		
	DBStd	8.22	4.51	5.04	5.23	4.62	4.41	3.26	2.78	2.68	3.42	4.48	5.03	4.66		
	HDD10.0	455	129	78	39	8	1	0	0	0	0	8	51	141		
	HDD18.3	1664	376	271	194	87	26	1	1	1	13	82	229	384		
	CDD10.0	2730	13	40	108	199	339	463	470	452	343	214	73	14		
	CDD18.3	898	0	1	6	28	106	214	212	194	106	30	1	0		
	CDH23.3	8001	0	3	51	254	1026	2222	1941	1590	720	186	7	0		
	CDH26.7	3232	0	0	4	51	394	1081	820	621	228	32	0	0		
(14) Wind	WSAvg	8.4	9.4	9.7	9.5	9.8	9.0	7.5	6.7	6.4	7.0	7.9	9.0	9.2		
(15) Precipitation	PrecAvg	302	11	12	11	11	17	28	56	57	54	27	14	15		
	PrecMax	584	55	45	33	50	84	121	115	139	136	79	74	72		
	PrecMin	186	0	0	0	0	0	0	13	1	0	0	0	0		
	PrecStd	99	16	11	8	14	21	26	31	34	42	23	18	19		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.9	23.4	26.9	30.6	34.7	38.0	36.3	35.3	33.2	29.5	24.3	19.2		
		MCWB	6.8	8.7	10.1	12.3	13.9	15.6	16.2	17.3	16.7	15.2	10.9	7.2		
	2%	DB	17.0	20.6	24.8	28.0	32.5	35.9	34.1	33.3	31.1	27.4	22.2	17.0		
		MCWB	6.3	7.6	9.4	11.0	13.2	15.0	16.6	17.5	15.9	13.4	10.0	7.0		
	5%	DB	15.0	18.3	22.6	26.3	30.9	34.2	32.6	31.8	29.1	25.5	19.8	14.7		
		MCWB	5.5	6.8	8.5	10.1	12.7	14.7	16.8	17.2	15.7	12.7	9.0	6.0		
	10%	DB	12.9	16.4	20.6	24.3	28.9	32.5	31.1	30.1	27.6	23.4	17.7	12.8		
		MCWB	4.6	6.0	7.7	9.5	11.9	14.6	16.6	17.0	15.6	12.1	8.3	5.0		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.8	10.3	12.4	14.5	16.8	19.0	20.3	20.1	19.7	17.7	13.5	9.9		
		MCDB	14.7	19.3	22.7	22.4	25.9	27.1	28.0	28.7	25.9	23.8	19.2	15.7		
	2%	WB	7.5	8.8	10.6	13.0	15.7	18.1	19.3	19.4	18.7	16.3	11.6	8.0		
		MCDB	14.1	17.4	21.2	22.4	24.8	26.9	27.3	28.1	25.2	21.9	18.4	14.4		
	5%	WB	6.4	7.7	9.5	11.7	14.8	17.5	18.7	18.8	17.9	15.1	10.3	6.7		
		MCDB	13.5	16.3	20.0	22.0	24.1	26.9	26.8	27.6	24.1	20.9	17.4	13.6		
	10%	WB	5.2	6.7	8.4	10.7	13.8	16.8	18.1	18.1	17.3	13.9	9.1	5.5		
		MCDB	12.4	15.1	19.1	21.6	23.4	26.3	26.2	26.6	23.3	20.0	16.3	12.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	10.6	11.5	12.4	12.9	12.6	12.8	11.1	10.9	10.5	10.9	10.6	10.1		
		MCDBR	12.1	13.0	13.9	14.2	14.2	14.3	12.8	12.5	12.6	12.3	12.2	11.8		
		MCWBR	6.4	6.3	6.3	6.2	5.7	4.7	3.9	3.9	4.4	5.2	5.6	6.1		
	5% WB	MCDBR	11.3	12.2	13.0	13.5	13.0	12.4	11.3	11.3	10.6	10.7	10.8	10.8		
		MCWBR	6.5	6.2	6.3	6.3	6.1	4.5	3.9	4.1	4.2	5.1	5.5	5.8		
(40) Clear-Sky Solar Irradiance	taub		0.234	0.241	0.254	0.280	0.304	0.317	0.345	0.329	0.314	0.271	0.253	0.239		
	taud		2.677	2.647	2.597	2.489	2.465	2.478	2.434	2.486	2.480	2.627	2.609	2.637		
	Ebn at Noon		1022	1038	1037	1011	981	961	933	948	956	987	986	997		
	Edh at Noon		72	82	93	108	112	110	115	107	103	83	76	71		
(44) All-Sky Solar Radiation	RadAvg		3.63	4.60	5.90	7.16	7.69	7.72	7.06	6.56	5.79	4.84	3.85	3.27		
	RadStd		0.30	0.37	0.34	0.40	0.38	0.35	0.36	0.29	0.34	0.41	0.31	0.23		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page